Rakesh Bal

Machine Learning Engineer · Energy & Infrastructure Forecasting · MLOps

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Summary

Machine learning engineer who builds production forecasting systems and the MLOps infrastructure that keeps them accurate and reliable, now applied to U.S. energy and electric grid infrastructure at Tapestry (X, Alphabet's moonshot factory). More than 4 years of full-time industry experience since a B.Tech from IIT Kharagpur, plus an M.S. in Computer Science from UCLA. Three peer-reviewed papers (ICWSM, AAAI, NeurIPS workshop) with 71 citations, and peer reviewer for IEEE ICRA and RO-MAN.

Work Experience

Google – X, The Moonshot Factory (Tapestry) **San Francisco, CA**
Software Engineer, Machine Learning Aug'26 - Present
- Building the GCP MLOps platform (*Vertex AI*, *Kubeflow Pipelines*) that trains, deploys & monitors electric grid forecasting models
- Standardizing configurable, Pydantic-validated training & evaluation DAGs so grid models are retrained and redeployed reliably
- Building reproducible CI/CD (GitHub Actions, immutable Docker digests) and fault-tolerant Pub/Sub monitoring & alerting

Stovell AI Systems **San Francisco, CA**
Machine Learning Operations Engineer Jan'24 - Jul'26
- Architected end-to-end training, deployment & monitoring pipelines for fuel price & demand forecasting across multiple customers
- Cut forecast MAE by ***15–30%*** via Transformer, LSTM & CNN hybrid architectures, time-series feature engineering & tuning
- Designed *causal* models for *price-volume elasticity*; built evaluation frameworks tying model performance to business ROI
- Ran production ETL & inference pipelines for multi-source data at a ***99.89%*** uptime SLA across all customer deployments
- Implemented data-quality validation, drift detection & accuracy tracking across geographies with *DataDog* and W&B
- Ran pre-deployment robustness testing, edge-case analysis and failure-mode identification for every model release

Kanini Software Solutions **Los Angeles, CA**
DevOps Engineer Aug'23 - Jan'24
- Automated provisioning of a healthcare platform with Terraform, improving scalability & reliability with AWS S3, Lambda & RDS

Amazon Web Services (AWS) **San Francisco, CA**
Applied Scientist Intern Jun'22 - Sep'22
- Benchmarked protein LLMs (ProtBERT/T5/GPT) and integrated diffusion docking for Drug–Target Interaction (DTI) prediction
- Cut training memory ***2x*** via data-parallel training on P4d.24xlarge (8× A100) with gradient checkpointing & mixed precision
- Improved baseline performance by ***20%*** and ***12%*** on two datasets with ProtBERT; project showcased at Amazon ML Conference
- Built an evaluation pipeline comparing models on accuracy, inference latency and compute cost

Goldman Sachs **Bangalore, India**
Software Engineer Aug'20 - Sep'21
- Led full-stack unification of two internal issue-tracking platforms into a single application and their migration to AWS
- Architected scalable RESTful APIs with Java & Spring Boot and an Angular & Redux UI; designed high-throughput schemas
- Owned the full SDLC with end-users through production rollout, driving adoption to ***50,000+*** daily active users

Research Experience

Scalable Analytics Institute (ScAI), UCLA **Los Angeles, CA**
Graduate Research Assistant | Advisor: Prof. Wei Wang Jan'23 - Sep'23
- Augmented DTI models with protein & molecule contact maps from diffusion docking and AlphaFold, a ***10%*** boost
- Explored cross-attention & contrastive-loss models for drug–protein interaction, leading to *PGraphDTA* (NeurIPS 2023 workshop)

Accenture Technology Labs & Indian Institute of Science (IISc) **Bangalore, India**
Research Intern | Advisors: K. M. Annervaz, Prof. Ambedkar Dukkipati May'19 - Aug'19
- Built a GCN stock-price model fusing news events and knowledge graphs, improving MSE by ***5%*** over baseline
- Used a knowledge graph as prior domain knowledge, with GCNs over events to model real-world market structure
- Implemented joint training of regression models and graph networks for more accurate time-series predictions

Indian Institute of Technology (IIT) Kharagpur **Kharagpur, India**
Undergraduate Researcher | Advisors: Prof. Mainack Mondal, Prof. Niloy Ganguly Oct'18 - Mar'20
- Built a cancer-misinformation dataset and BiLSTM-CRF detection model for Twitter; published at *ICWSM 2020*
- Benchmarked health-text embeddings (TF-IDF-weighted PubMed best); analyzed spread across cancer prevention, cure & causes
- Measured two-sided recommender fairness with electoral methods (SNTV, k-Borda, Chamberlin–Courant); *AAAI 2021*

Other Research Projects

Algorithmic Bias Discovery & Mitigation | Advisor: Prof. Niloy Ganguly — Apr'18 - Sep'18
– Detected bias in the COMPAS recidivism dataset with deep networks; proposed mitigation via Layer-wise Relevance Propagation
Overcoming Dataset Bias with GAN Augmentation | Advisor: Prof. Abir Das — Feb'19 - Apr'19
– Balanced gender representation in image datasets through unsupervised domain translation with CycleGAN

Education

University of California Los Angeles (UCLA) — **Los Angeles, CA**
Master of Science (M.S.) in Computer Science | GPA - 3.81/4.0 — Sep'21 - Jun'23
Indian Institute of Technology (IIT) Kharagpur — **Kharagpur, India**
Bachelor of Technology (B.Tech.) in Computer Science and Engineering | GPA - 9.04/10.0 — Jul'16 - May'20

Relevant Coursework

Graduate (UCLA): Theoretical ML, NLP, Advanced Data Mining, Graph Neural Networks, Bayesian Networks, Human-Centered AI
Undergraduate (IIT KGP): Algorithms, Probability & Statistics, ML, Deep Learning, Reinforcement Learning, AI & Ethics

Publications

PGraphDTA: Improving Drug–Target Interaction Prediction using Protein Language Models & Contact Maps [Paper]
– **Rakesh Bal**, Yijia Xiao, Wei Wang
– *NeurIPS 2023 Workshop on AI for Science* | *15 citations* | First author
– Cited in 2025 review articles in *Computers in Biology and Medicine* and *Current Opinion in Structural Biology*
Two-Sided Fairness in Non-Personalised Recommendations [Paper]
– Aadi Swadipto Mondal*, **Rakesh Bal***, Sayan Sinha*, Gourab K. Patro
– *AAAI 2021, Student Abstract & Poster Program* | *10 citations* | Co-first author
Analysing the Extent of Misinformation in Cancer Related Tweets [Paper]
– **Rakesh Bal***, Sayan Sinha*, Swastika Dutta, Rishabh Joshi, Sayan Ghosh, Ritam Dutt
– *ICWSM 2020* | *46 citations* | Co-first author
– Independently cited by Stanford (IEEE S&P, ICWSM), Oxford (*Cancer Discovery*), Cornell (*JMIR*), UIUC, Max Planck (CSCW)
* equal contribution · citation counts from Google Scholar, November 2026

Service & Teaching

Peer Reviewer: IEEE International Conference on Robotics and Automation (ICRA) 2025; IEEE RO-MAN 2025
Teaching: UCLA Graduate Teaching Associate, CS32 (Intro to CS II) & Chem 20A, 5 quarters, 500+ students, 500+ hours
National Digital Library of India: Prepared worked solutions to past IIT-JEE papers for the NDL's public JEE Archive

Projects

DeepSub: Fine-tuning LLMs for multi-source subtitle translation [Code]
– Fine-tuned Gemma & LLaMA with QLoRA to fuse context from 2 source languages and translate subtitles to a target language
– Evaluated a triplet translation pipeline and experimented with 6 *Indic* & 3 *European* languages with BLEU performance tracking
Autonomous Underwater Vehicle (AUV), IIT Kharagpur [Link]
– Engineered real-time sensor data collection and processing pipeline on an Autonomous Underwater Vehicle using CVAT & ROS
– Built real-time buoy detection (SSD-MobileNet) on camera & IMU streams, optimized for an on-board Movidius NCS
– Used ROS packages like Actionlib and Smach to implement the Mission Planner Stack, participated in SAVe at NIOT, Chennai

Skills

Languages: Python, Pydantic, Java, C, C++, Go, SQL, JavaScript, TypeScript, CUDA
ML & MLOps: PyTorch, TensorFlow, JAX, vLLM, TensorRT, Vertex AI, Kubeflow, MLflow, DataDog, Snowflake
Cloud & Web: Docker, K8s, GCP, AWS, Terraform, Angular, React, Flask, Node.js, MySQL, GraphQL, Spring Boot, MongoDB

Achievements

– Amazon ML Conference 2022: AWS drug–target interaction project selected for showcase at Amazon's internal ML conference
– IIT-JEE Advanced 2016: All India Rank 265/200K, top 0.15% of candidates for admission to the IITs
– IIT-JEE Mains 2016: All India Rank 187/1.3M, top 0.02% in India's national engineering entrance exam
– KVPY Fellowship: All India Rank 191/100K, Govt. of India research fellowship administered by IISc Bangalore
– Puri District Chess Cup: 5th/1000 players in an open district-level chess championship